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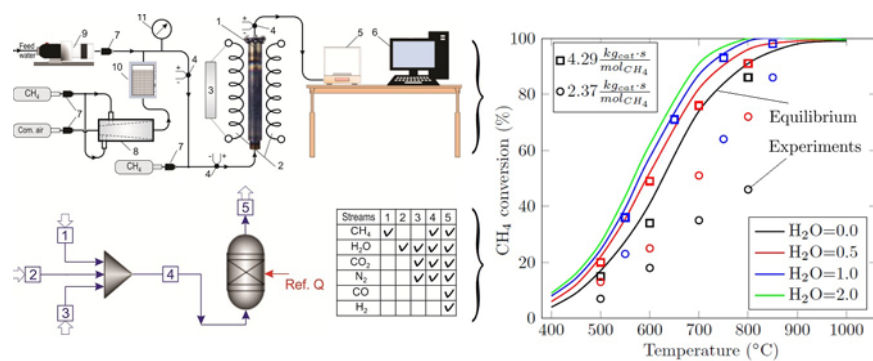
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Combined methane reforming with a mixture of methane combustion products and steam over a Ni-based catalyst: an experimental and thermodynamic study

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Abstract

Combined methane reforming with a mixture of methane combustion products and steam over a commercial Ni-based catalyst is studied both experimentally and theoretically. The 7-holes cylinder of NiO- α -Al₂O₃ catalysts is used as the pellets. The thermodynamic analysis of the reforming process was carried out by total Gibbs free energy minimization to determine the maximum achievable methane conversion in the reformer. The reforming performances were determined for various amount of extra steam in inlet reaction mixture (from 0 to 2 mole of H₂O per mole of CH₄). The methane conversion is increasing when steam is added to the reaction mixture at a fixed temperature. The methane conversion reaches a maximum value (about 1) at the temperature range above 800°C. Based on experimental results, the approximation functions of the methane conversion versus the residence time for the various temperatures and methane-to-oxidizer ratio were obtained. The methane conversion reaches the value that is close to equilibrium at the residence time above 4.0 kg_{cat}·s/mol. The effect of pressure (in the pressure range from 1 to 5 bar) on methane conversion is insignificant and is no more than 1-2%. The values of the methane conversion can be used for the energy analysis of the thermochemical recuperation system by methane reforming with steam and flue gases.

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1. Introduction

Reforming of hydrocarbon fuels is the main method for hydrogen, carbon monoxide and synthesis gas production. Among various hydrocarbons, natural gas is used most often for reforming [1–6]. Nowadays, approximately 90-95% of the hydrogen produced in all the world is made by natural gas reforming [1]. Moreover, the reforming processes for hydrogen-rich gas are currently the subject of R&D, involving the reforming of other fuels such as methanol, propane, diesel fuel, and ethanol [7–11]. Reforming has a wide application in the chemical industry, but in recent years the use of reforming has become increasingly used in the energy industry [12, 13], transport [7, 14–16], and other industries. The use of methane reforming has found wide application in the energy sector in the thermochemical waste-heat recuperation (TCR) systems [17–20]. In such systems, endothermic reactions of methane reforming are used to produce a new fuel for fuel-consuming equipment. TCR is based on two fundamental principles. The first is the increase in the LHV of the synthetic fuel compared to the original fuel; and second is the transformation of the exhaust heat into chemical energy of the new synthetic fuel.

Main endothermic reactions of methane reforming are steam (SMR) and dry (DMR) reforming, as well as combined steam-dry reforming. SMR and DMR are usually performed at temperatures up to 1000°C [19, 21–23]. In addition, in order to obtain a high methane conversion, the oxidizer is supplied to the reformer more than the stoichiometric amount, for example, $\text{H}_2\text{O}:\text{CH}_4=2\dots3$ [24–26]. In light of this, it's becoming apparent that the oxidizer production and heating to high temperatures requires a large amount of heat. For example, for the production and heating of steam at a ratio $\text{H}_2\text{O}:\text{CH}_4=2$ and $T=1000\text{K}$, about 10% of all supplied heat for SMR is required [17, 27]. One way to solve these problems (an oxidizer production and heating) is to use high-temperature

combustion products for methane reforming. In particular, the products of the complete combustion of methane with an excess air coefficient of 1.0 consist mainly of nitrogen (N_2), carbon dioxide (CO_2) and steam (H_2O). Moreover, the mole fraction of oxidizers (H_2O+CO_2) in the combustion products is about 30%.

The use of combustion products as an oxidizer for methane reforming has found application in the thermochemical waste-heat recuperation systems [28–30]. This method of methane reforming has attracted interest from both energetic and environmental perspectives. Firstly, the combustion products, for example, after high-temperature plants have a high temperature (up to 800–1000°C [29]). Secondly, from an environmental point of view, one of the most abundant carbon containing greenhouse gas - carbon dioxide, can be utilized effectively in this process and converted into useful chemical products. There are publications in the literature devoted to the use of methane reforming with the products of methane complete combustion [31].

Gaber and co-workers experimentally investigated of a thermochemical regeneration waste heat recovery process using a reformer unit [29]. They determined the synthesis gas composition after methane reforming with the flue gases over a commercial Ni-based catalyst. The obtained results show the efficiency of using this a methane reforming method. A methane conversion rate between 51.2 and 95.3% was achieved under different gas hourly space velocities. However, the authors considered a small number of practically interesting cases for the use of methane reforming with the flue gases in the TCR systems. In addition, in this paper the concerns of the effect of steam addition on the steam methane reforming process were not considered.

Lemonidou and Vasalos reported the results of experimental investigation of methane reforming by carbon dioxide with steam addition over 5 wt.% Ni/CaO- Al_2O_3 catalyst [32]. The authors determined that steam addition to carbon dioxide has a meaningful influence on the methane reforming process. When dry methane reforming is performed contemporaneously with SMR, a significant increase in the methane conversion and a drastic decrease in carbon deposition

takes place. The various inlet gas compositions were used for the experimental investigation. In this work [32], researchers used pure oxidizers such as steam and carbon dioxide for the methane reforming process, while the real flue gases also contain nitrogen. Moreover, in this investigation the mixture of methane and oxidisers was heated before the reforming process from about 200 °C to reforming temperature.

The works mentioned above, as well as a series of other articles [7, 15, 20], show the promise of further research of the methane reforming process with the combustion products. Moreover, for the high-temperature plants, it is of interest to use TCR by methane reforming with a mixture of combustion products and steam. Therefore, it is necessary to further investigate the effect of steam addition to the combustion products for the methane reforming process.

The objective of this work is to determine an effect of operational parameters on the methane reforming process when a hot mixture (mixture temperature equal to reforming temperature) of flue gases and steam is used as an oxidiser. An experimental investigation of methane reforming over a commercial Ni-based catalyst is performed. Thermodynamics equilibrium analysis is conducted to determine the maximum achievable methane conversion and the effects of operational parameters such as temperature and inlet reaction mixture composition on the equilibrium synthesis gas composition at equilibrium conditions. The results of the experiments and thermodynamic analysis are compared to evaluate the effect of residence time, temperature, pressure, and inlet gas composition on the methane reforming efficiency. The products of methane combustion obtained by methane combustion with an excess air coefficient of 1.0 in a tangential burner are used for methane reforming. The main intention of this work is to investigate the maximum number of practically interesting cases in order to develop the TCR systems. In addition, the results of this work can be useful for researchers in the chemical industry and metallurgy.

2. Experiments

2.1. Experimental setup

A schematic diagram of the experimental setup employed during the present research is shown in Fig. 1. The experimental setup can be virtually divided into three blocks: (1) the block for preparing of the reaction mixture (element numbers – 7, 8, 9, 10); (2) the block of the reforming process (element numbers – 2, 3); (3) the block for measuring and storing of the results (element numbers – 5, 6). The main elements of the first block are an electrical steam generator and a gas burner. The main element of the second block is a methane reformer. The main component of the third block is a portable gas analyzer.

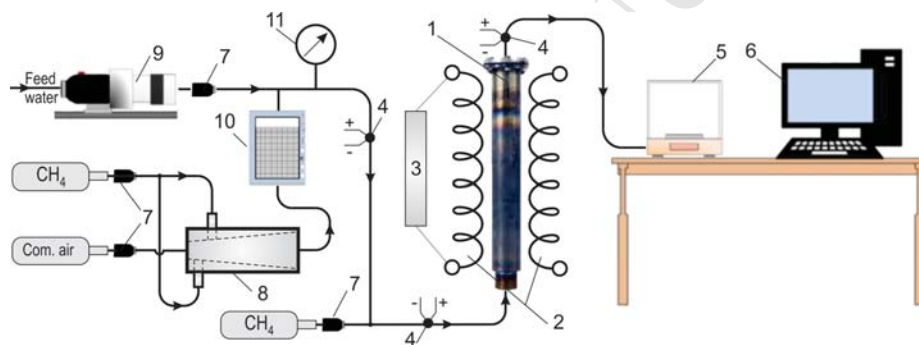


Figure 1: Schematic view of experimental setup

The experimental setup mainly consists of the following elements: 1 - a methane reformer; 2 - an electric muffle furnace; 3 - a watt-hour meter; 4 - the thermocouples; 5 - a portable gas analyzer; 6 - a personal computer; 7 - the mass-flow meters; 8 - a tangential burner; 9 - an electrical steam generator; 10 - a gas intercooler; 11 - a manometer.

The methane reformer made of high-alloy steel has been placed in an electric muffle furnace. The electrical furnace heats the reaction space up to 900°. The thermocouples are made of a tungsten-rhenium alloy. The gas composition is controlled by the portable gas analyzer Gasboard 3100, which can be connected to a personal computer. In a leakproof gas burner with a tangential admission

of methane, methane combustion takes place with an excess air factor equal to unity. Superheated steam is produced in the electric steam generator PG-LEK 1.1 (in Russian). High purity methane is used as a fuel and as a reagent for reaction reforming in the experiments, as well as a high purity compressed air is used for combustion. High purity distilled water is used for a steam production.

A photograph of the experimental setup and some elements of the schematic diagram are shown in Fig.2. The element numbers in Fig.2 coincide with the figure caption to Fig.1.

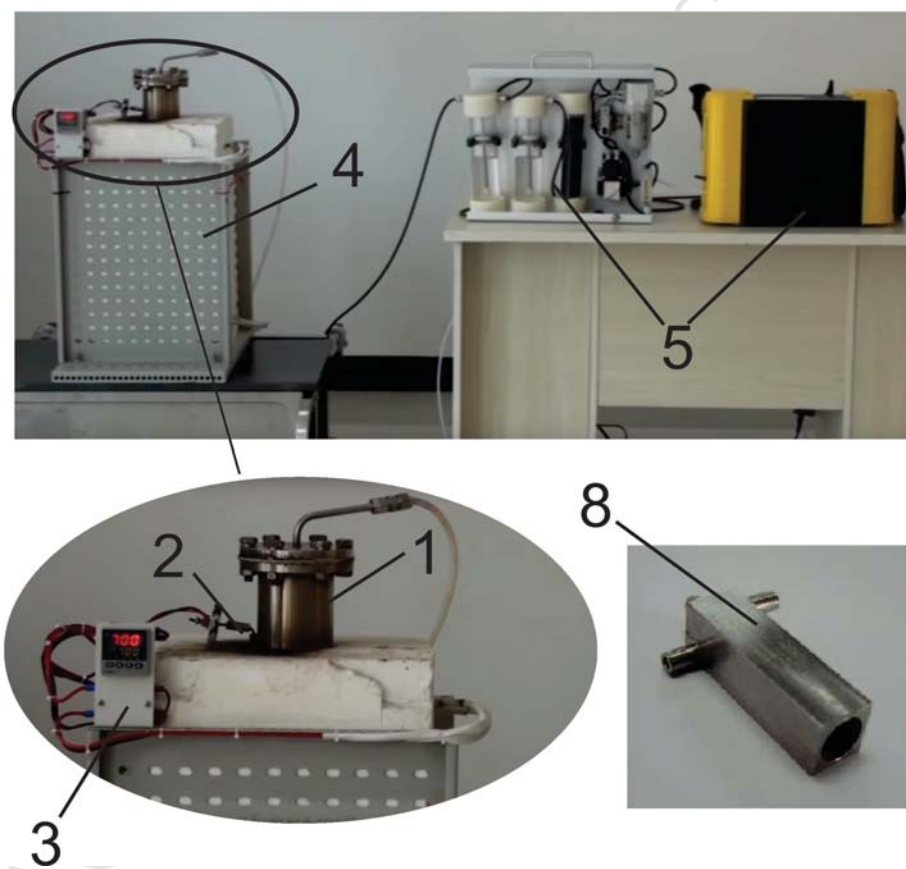


Figure 2: Photograph of experimental setup

2.2. Catalyst

The reaction space of the methane reformer is filled with the commercial Ni-Al₂O₃ catalyst produced by NIAP-CATALYZATOR LLC. For experiment the Ni-Al₂O₃ catalyst is used with the following chemical composition: NiO = 14.5%; SiO₂ = 0.2%; support CaO-MgO-La₂O₃-Al₂O₃ [22]. Other properties are presented below: bulk density is 680 kg/m³; surface area is 4.5 m²/g_{cat}; the average porosity of catalyst particles is about 41%. One type of the particle shapes is used for the experiments – the cylinders with 7 holes.

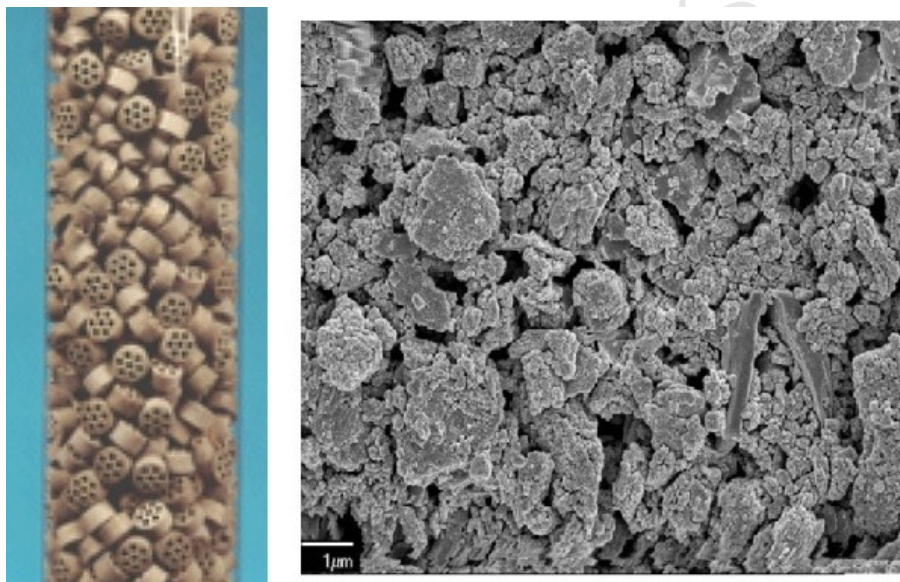


Figure 3: Left side are photographs of the packed beds filled by the catalysts. Right side is SEM image of fresh Ni-Al₂O₃ catalyst provided by NIAP-CATALYZATOR LLC.

The catalyst particles are placed into the tube with a diameter of 100 mm. The catalysts layer depth is 600 mm. The geometrical characteristics of catalysts are following: diameter – 25 mm, thickness – 25 mm, holes – 7x5 mm.

SEM image of the catalyst is presented in Fig.3 (right side). Scanning Electron Microscopy (SEM) photo of catalyst structure before experiments (fresh catalyst) is provided by the catalyst manufacturer NIAP-CATALYZATOR LLC (Novomoskovsk, Russia).

2.3. Experimental procedure

In the present study, a set of experiments was performed for various operational parameters: a temperature range from 400 to 800°C, a pressure from 1 to 5 bar, a composition of the inlet reaction mixture, a residence time. The residence time was varied due to a change in the flow rate of the reaction mixture and inlet reaction mixture composition. Moreover, a set of experiments was performed when steam is added to the flue gases.

Before the experiment, the electric furnace heats the catalyst to the predetermined temperature. When the catalyst is heated, the reaction space of the methane reformer is filled with nitrogen. Nitrogen is used to obtain the same temperature in each part of the reaction space, and to safety preheat of the reaction space. Heating of the catalyst takes approximately 25-27 minutes. During this period, the temperature in the reformer stabilizes throughout the reaction space. In parallel with the heating of the catalyst, combustion of methane is regulated by using of flowmeters. The temperature of the combustion products is regulated in the gas intercooler to a temperature that is higher than the catalyst temperature in the reformer. This is due to the fact that when the methane for reforming reaction is fed into the combustion products, the temperature of the mixture will decrease. The target temperature of the flue gases after the burner is determined experimentally in such a way that the temperature of the reaction mixture before the reformer is equal to the temperature of the reaction space (catalyst) of the reformer.

At the initial time, the system of flue gas supply to the reaction space is not connected to the reformer. After the reformer becomes warmed up and the combustion of methane is regulated, the flue gas supply system is connected to the reformer. With the help of an electric heater, the temperature in the reaction space during the experiments is regulated. Each experiment has a duration of 10-15 minutes. For the analysis of catalytic activity, few experiments have a duration up to 12 hours. The fresh catalysts are used for each experiment. During the experiment, the composition of the reformed products and the heat consumption for the reforming reaction are fixed every 5 minutes.

For this investigation, the inlet gas compositions were controlled by the mass flowmeter. Number of moles for each component are determined by calculation by the expression:

$$n_i = \frac{G_i}{\mu_i}, \quad (1)$$

where G_i is mass flow-rate of i component, g/s; μ_i is molal mass of i component, g/mol.

The outlet products were controlled by the gas analyzer Gasboard 3100. The portable gas analyzer was connected to laptop to display the composition of syngas. The calibration procedure of Gasboard 3100 was made by injection of a gas mixture with a predetermined composition. The calibration procedure was repeated every 10 experiments.

3. Thermodynamic analysis

The algorithm of the thermodynamic analysis by the Gibbs free energy minimization (GFEM) was introduced in the author's previous publications [13, 33, 34] and it was also described in details by other authors [35–37]. Thermodynamic analysis of combined methane reforming with products of methane combustion and steam via GFEM method was introduced to determine equilibrium compositions of the product gas. The GFEM technique is based on the principles that the analyzed chemical system is thermodynamically auspicious if total Gibbs free energy has the lowest value and differential of Gibbs free energy is zero for presented operation conditions [38]:

$$dG^t = 0 \quad (2)$$

where G^t is total Gibbs free energy.

At moderate low pressure (before 5-10 bar) and high temperature, the system can be considered as ideal [39]. Moreover, the reaction products from the reformer are determined, based on the following admissions: steady flow process with the negligible change of kinetic and potential energies; perfect insulation

of reformer with negligible heat loss; chemical reactions with known reactants and products compositions are equilibrium.

Total Gibbs free energy of a methane reforming system is presented by the sum of the chemical potential of all chemical elements:

$$G^t = \sum n_i \mu_i \quad (3)$$

where n_i is number of moles of species i ; μ_i is chemical potential of species i .

The chemical potential of each element is determined as follow:

$$\mu_i = \Delta G_{fi}^0 + RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (4)$$

where G_{fi}^0 is standard Gibbs free energy of formation of species i ; R is molar gas constant; T is temperature of the system; f_i is fugacity of pure species i ; f_i^0 is fugacity of pure species i at the standard state.

If combine the expressions above, the equation for total Gibbs free energy of the chemical system can be written as:

$$G^t = \sum n_i \Delta G_{fi}^0 + \sum n_i RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (5)$$

The thermodynamic analysis was performed by Aspen-HYSYS version 8.4, which is a powerful software package for modeling unsteady-state and steady-state chemical process for different systems [40–43]. The calculation model in Aspen HYSYS is presented in Fig.4.

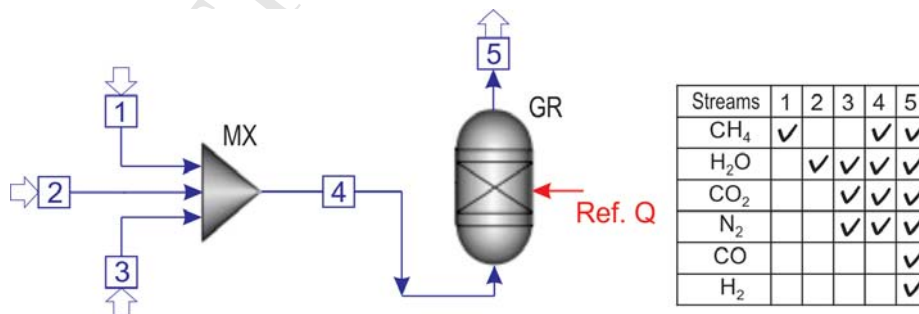


Figure 4: The calculation model of the methane reforming process: MX – a gas mixer; GR – a Gibbs reactor; Ref. Q – heat for the reforming process.

The overall reaction in the methane reformer to form H_2 , CO , CO_2 , H_2O , and other components may include the equations listed in Table 1 [24]. In addition, other chemical reactions may occur in the reformer, which is basically a combination of the reactions shown in the Table 1. However, it is well known that for the thermodynamic analysis it is sufficient to consider only elementary reactions presented in Table 1.

Table 1: Possible chemical reactions in the reformer [25]

No.	Reaction	ΔH_{298} , (kJ/mol)
Methane oxidation		
1	$CH_4 + H_2O \rightleftharpoons CO + 3H_2$	206.4
2	$CO + H_2O \rightleftharpoons CO_2 + H_2$	-41.1
3	$CH_4 + CO_2 \rightleftharpoons 2H_2 + 2CO$	247.3
Carbon formation		
4	$CH_4 \rightleftharpoons C + 2H_2$	74.8
5	$2CO \rightleftharpoons C + CO_2$	-173.3
6	$CO + H_2 \rightleftharpoons C + H_2O$	-90.13
7	$CO_2 + 2H_2 \rightleftharpoons C + 2H_2O$	-131.3

Fig.5 shows the dependence of the Gibbs free energy versus temperature for reactions presented in Table 1. The standard equation to obtain results for Fig.5 can be written as following:

$$\Delta G^0 = R \cdot T \ln K_{eq}, \quad (6)$$

where K_{eq} is equilibrium constant. The data for K_{eq} were obtained from Aspen HYSYS database.

Equation (6) allows us to determine the change in Gibbs free energy for a chemical reaction from Table 1 in an equilibrium state. The equilibrium constants of the reactions from Table 1 are well known. For different databases, the values of equilibrium constants may differ slightly. In this article, reaction equilibrium constants are taken from the IVTANTHERMO database [44] developed by Joint Institute for High Temperatures of the Russian Academy of Sciences.

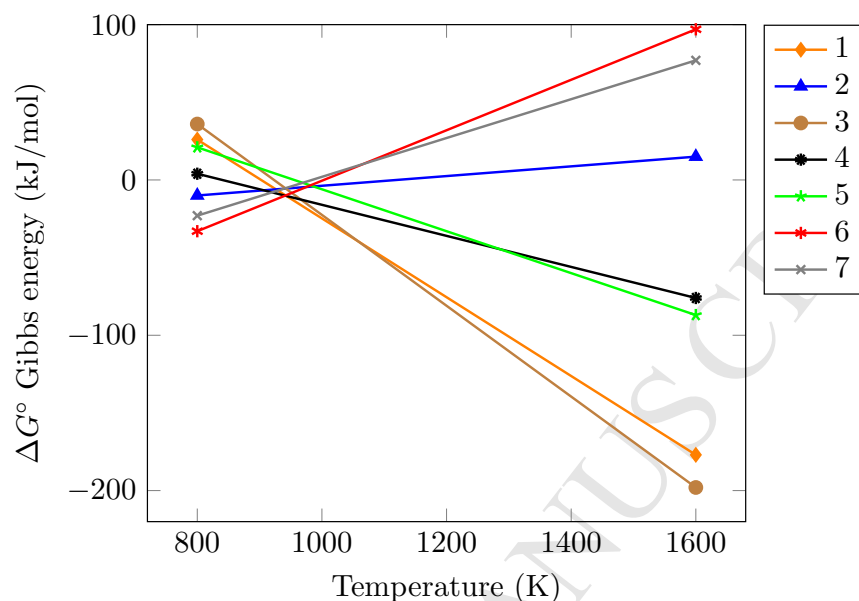


Figure 5: The Gibbs free energy as a function of temperature for reactions in Table 1

There are many other databases, such as FactSage [45], etc, that directly calculate the change in Gibbs free energy. I used IVTANTHERMO because my institute is licensed to use such database.

As it can be seen from Fig.5, the Gibbs free energy for oxidation reactions (1 and 3 from Table 1) decreases significantly with increasing temperature. In the temperature range above 900K, reforming reactions (1 and 3 from Table 1) are more likely to be thermodynamic than the reactions of methane cracking (4) and Boudouard reaction (5) [46]. At temperatures above 1000K, the equilibrium of the reactions (6 and 7) is shifted toward the formation of gasification products of the carbon. Therefore, it can be concluded that the reactions of steam and carbon dioxide reforming mainly take place in the methane reformer. However, in the experiment, traces of soot carbon on the catalytic surface were found at a methane-to-oxidizer ratio of about 1. This fact is discussed below.

4. Results and discussion

To analyze the effectiveness of the methane reforming process, the methane conversion are defined as follows:

$$X_{CH_4} = \frac{CH_{4(in)} - CH_{4(out)}}{CH_{4(in)}}. \quad (7)$$

4.1. Effect of temperature

Fig.6 shows changes of CH_4 conversion as a function of temperature when the various components of inlet reaction mixture are fed into the methane reformer at the pressure of 1 bar for the different residence times.

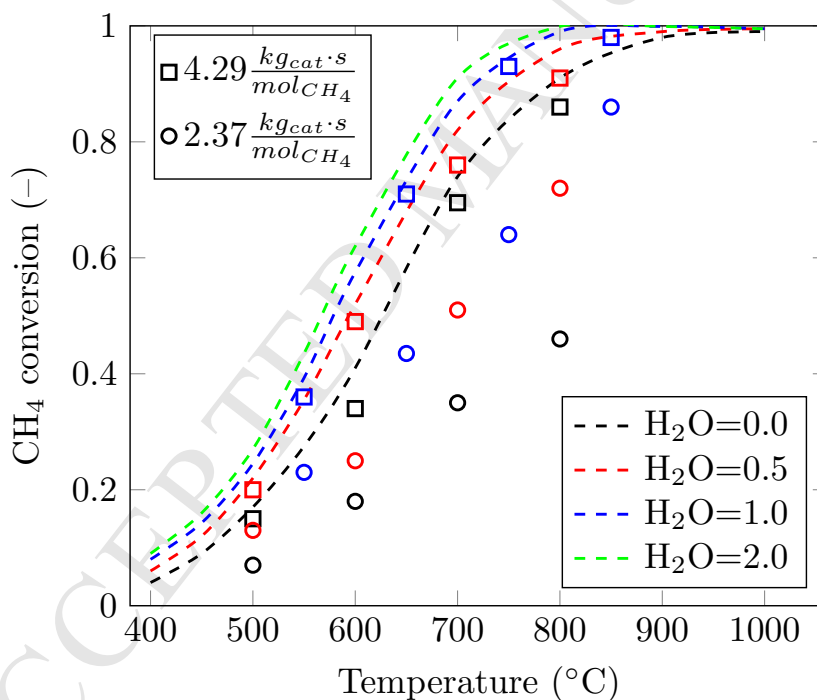


Figure 6: Effect of temperature on CH_4 conversions at $p=1\text{bar}$ for the various inlet gas compositions: dashed line – equilibrium condition; hollow point – experimental results

Fig.6 depicts the experimental data and the results of the thermodynamic analysis. Here, the equilibrium CH_4 conversion were compared with the CH_4

conversion that obtained for the various residence times. The experiments are performed for different residence times: $4.29 \text{ kg}_{cat} \cdot \text{s/mol}$ and $2.37 \text{ kg}_{cat} \cdot \text{s/mol}$. When the experimental conditions changed (steam addition and temperature), the residence time did not change more than 2.0%. The experiments were carried out at a fixed flow-rate of flue gases ($\text{CH}_4:\text{FG}=\text{const}$). The stoichiometric amount of oxidants (H_2+CO_2) in the flue gases is chosen as a fixed value ($\text{CH}_4:\text{CO}_2:\text{H}_2\text{O}:\text{N}_2=1:0.33:0.67:2.51$). With increasing the residence time, the syngas composition approaches to equilibrium composition (dotted lines). With an infinitely long time, it is assumed that the syngas will reach an equilibrium composition when concentrations of each elements in syngas have no further tendency to change with time. In the Fig.6 is clear that, for low mole flow rates, hence high residence time, the methane conversion is higher than it for low residence time. The increase in mole flow rate leads to a decrease in the residence time, therefore the number of moles passing through the reaction space increases and the methane conversion decreases.

Fig.6 shows the positive effect of temperature on CH_4 conversion. As the temperature increases, the methane conversion increases for each residence time. When the residence time is low, the methane conversion is almost proportional to temperature. However, the effect of temperature on CH_4 conversion has a non-linear relationship at the high residence time. Moreover, the steam addition to the feed stream increased the methane conversion considerably by 20-71% for the different temperatures. For a high residence time at the presence of additional steam, CH_4 conversion tends to an equilibrium value. In addition, Fig.6 shows that an additional increase of steam amount above 2.0 gives a slight increase in conversion. For the thermochemical recuperation systems, the determination of the optimum amount of steam is an important task because additional heat is expended on the production and heating of the steam [13].

For all experiments, the standard deviation of the methane conversion was determined. The inlet gas composition varies by steam addition. The initial inlet gas composition is $\text{CH}_4:\text{CO}_2:\text{H}_2\text{O}:\text{N}_2=1:0.33:0.67:2.51$. Standard deviations for experimental measurements of the methane conversion are: 0.008 for

$H_2O=0.0$; 0.018 for $H_2O=0.5$; 0.024 for $H_2O=1.0$; 0.032 for $H_2O=2.0$. The relatively large error with increasing steam addition can be explained due to the fact that with increasing steam addition the mole fraction of unreacted methane decreases. This result is not surprising since the standard deviation is increasing due to the decrease in the denominator in an expression for standard deviation.

4.2. Effect of residence time

For more detailed analysis of the methane reforming process, two parameters are introduced: residence time for weight and residence time for surface. First (residence time for weight W_{cat}/F_{CH_4}) is calculated as catalyst pellet weight divided by the feed rate of the methane and expressed as $kg_{cat}s/mol$. The weight residence time shows a mass of catalyst bed per mole flow-rate on methane inlet. Second (residence time for surface S_{cat}/F_{CH_4}) is calculated as a surface of a packed bed divided by the feed rate of the methane and expressed as m_{cat}^2s/mol .

Fig.7 depicts the CH_4 conversion as a function of the residence time at different temperatures and pressure 1 bar at a fixed methane-to-oxidizer ratio ($CH_4:CO_2:H_2O:N_2=1:0.33:0.67:2.51$). Dashed lines denote the equilibrium methane conversion for each temperature. It is seen that at the fixed temperatures, CH_4 conversion increases with increasing residence time. However, CH_4 conversion increases slightly in the temperature range above $700^\circ C$ when high residence time is taking place. Obviously, with a further increase in the residence time, CH_4 conversion reaches a maximum and stable value because the methane reforming process reaches to close to the equilibrium condition.

For engineering calculations of the methane reforming process, it is convenient to use expressions in which the methane conversion depends on the residence time. Third-degree polynomial regressions were used for correlating relationships between CH_4 conversion and residence time. The relationships for the experiments of methane reforming at a fixed temperature, pressure and the

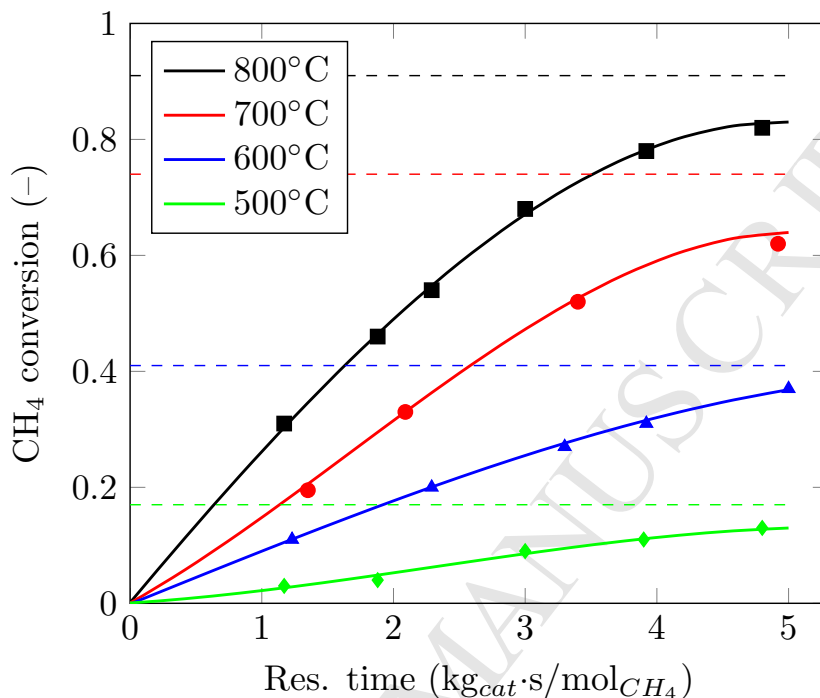


Figure 7: Methane conversion as a function of weight residence time at different temperature for $\text{CH}_4:\text{CO}_2:\text{H}_2\text{O}:\text{N}_2=1:0.33:0.67:2.51$ (no added steam $\text{H}_2\text{O}_{add}=0$). Here: dashed line – equilibrium conditions; solid line – approximating curve; solid mark – experimental results

ratio of methane-to-oxidizer were obtained as follows:

$$X_{\text{CH}_4} = a_0 + a_1 \left(\frac{W}{F_{\text{CH}_4}} \right) + a_2 \left(\frac{W}{F_{\text{CH}_4}} \right)^2 + a_3 \left(\frac{W}{F_{\text{CH}_4}} \right)^3, \quad (8)$$

where a_i – correlation coefficients; W – mass of catalyst, kg; F_{CH_4} – molar flow rate of CH_4 , mol/s.

The results of the regression analysis are presented in Table 2.

Fig.8 shows the variation of the total methane conversion versus residence time at a fixed temperature of 600°C and fixed $\text{CH}_4:\text{CO}_2:\text{H}_2\text{O}:\text{N}_2=1:0.33:0.67:2.51$ ratio. The amount of steam varies from 0.0 to 2.0 mole for Fig.8. It is seen that at a fixed reforming temperature, the total methane conversion increases with increasing residence time and the addition of extra steam.

To assess the accuracy of the approximation expressions in Table 2, an anal-

Table 2: Correlation coefficients for Eq.8

	T, °C	a ₀	a ₁	a ₂	a ₃
H ₂ O=0.0 p=1bar	800	0.00126243	0.27143	-0.00851387	-0.00252493
	700	0.000590199	0.127593	0.0247295	-0.00493837
	600	-0.00128687	0.0919071	-0.0000197753	-0.000714569
H ₂ O=0.5 p=1bar	800	0.00047619	0.325635	-0.0509524	0.00388889
	700	-0.00404762	0.223056	-0.00511905	-0.00222222
	600	0.00325397	0.138545	-0.0113294	0.0000462963
H ₂ O=1.0 p=1bar	800	-0.00111111	0.350503	-0.0488492	0.00287037
	700	-0.000793651	0.235529	-0.0112698	-0.000925926
	600	-0.000238095	0.174563	-0.0238095	0.00138889
H ₂ O=2.0 p=1bar	800	0.00238095	0.519127	-0.105476	0.007777778
	700	0.00015873	0.1366085	-0.0488889	0.00185185
	600	0.00201681	0.189146	-0.0160084	-0.000196078

ysis of the error distribution is performed. A histogram of the error distribution of the approximating expressions is shown in Fig.9. Fig.9 shows that the most accurate expressions are obtained for the stoichiometric methane-to-oxidizers ratio $\text{CH}_4:\text{CO}_2:\text{H}_2\text{O}:\text{N}_2=1:0.33:0.67:2.51$.

The statistical analysis by the Pearson criterion, performed on the basis of the 9-bit histogram in Fig.9, showed that the actual error distribution of the approximating function for CH_4 , $p=1\text{bar}$, cylinder, $T=700^\circ\text{C}$ within the residence time from 0 to $4 \text{ kg}_{cat}\cdot\text{s}/\text{mol}$ is in good agreement with the law of the normal Gaussian distribution ($\chi_{func} = 4.09 < \chi_{exp} = 12.592$, with a confidence level of $P=95\%$ and the number of degrees of freedom $m=6$). Therefore, for engineering calculations with high accuracy, it is possible to use approximation dependences, given in Table 2. The most accurate expressions are obtained for the stoichiometric methane-to-oxidizers ratio. The decrease in accuracy when adding steam can be caused by the fact that additional elements appear in the schematic diagram in Fig.1.

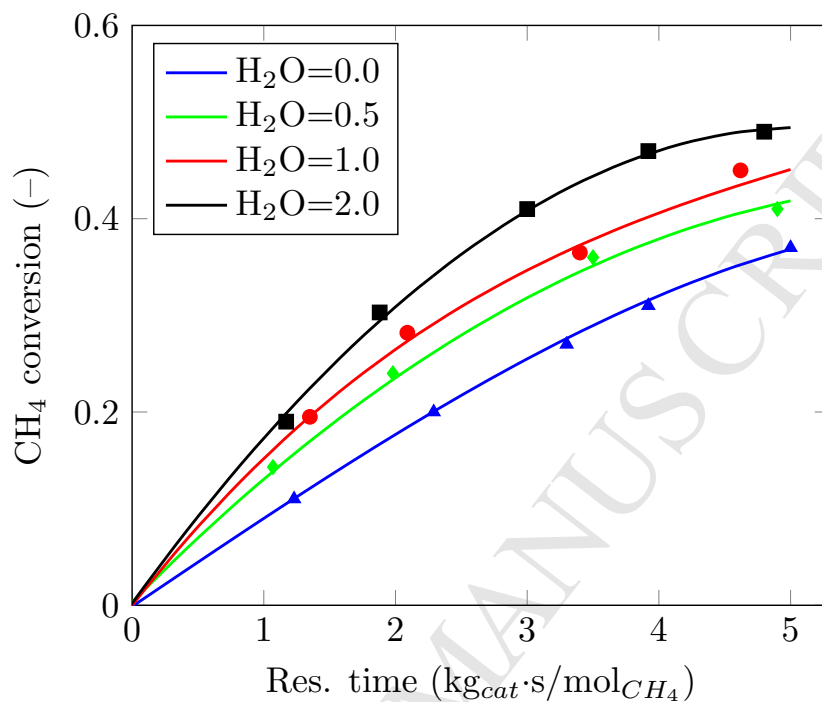


Figure 8: Methane conversion as a function of residence time at fixed temperature 600 °C, fixed $\text{CH}_4:\text{CO}_2:\text{H}_2\text{O}:\text{N}_2=1:0.33:0.67:2.51$ ratio and $p=1\text{bar}$: solid line – approximating curve; solid mark – experimental results

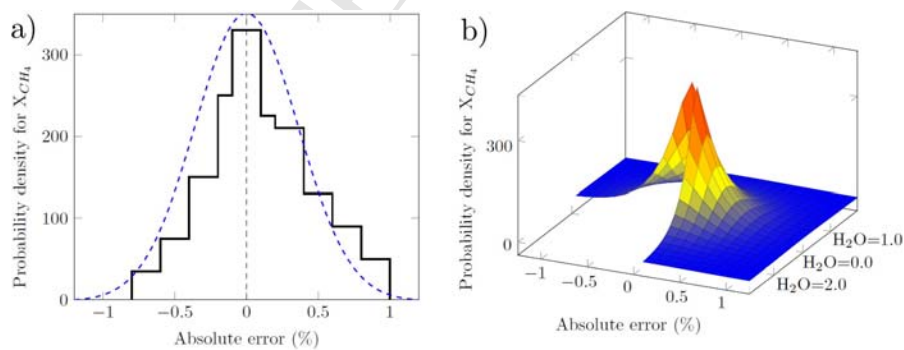


Figure 9: Histogram of error distribution of the approximating functions: a) $T=700^\circ\text{C}$, $\text{H}_2\text{O}=0.0$ (dashed line – Gaussian curve; solid line – histogram); b) Fixed $T=700^\circ\text{C}$ for $\text{H}_2\text{O}=0.0\dots2.0$.

Fig.10 shows plots of the residuals for some approximation expressions from Table 2. Fig.10 shows the difference between the experimental value and the predicted value by the approximation expressions. Fig.10 shows that the differences in the CH₄ conversion values for all the investigated compositions are minimal.

$$Residual = Res.time^{experiment} - Res.time^{approxim} \quad (9)$$

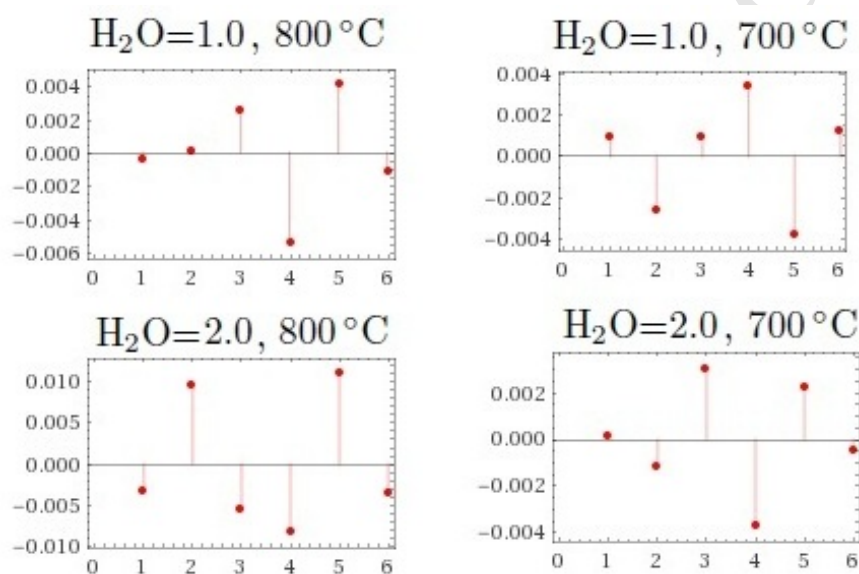


Figure 10: Plots of the residuals

The data from Table 2 are important for the engendering calculations of the thermochemical waste-heat recuperation (TCR) systems. As it is known, the main concept of the thermochemical recuperation system is the transformation of exhaust gases heat into chemical energy of a new synthetic fuel that has higher calorimetric properties such as low-heating value. Therefore, the efficiency of the TCR system depends on the enthalpy of the methane reforming process. Based on the heat balance equation for the methane reformer that presented in Fig.1, the enthalpy of the overall methane reforming process can be written as:

$$H_{overall} = X_{CH_4} \cdot H_{mr} + X_{CO_2} \cdot H_{wgs}. \quad (10)$$

where H_{mr} – the enthalpy of the combined steam-dry methane reforming reaction; X_{CO_2} – the methane conversion to carbon dioxide via water-gas shift reaction $X_{CO_2} = CO_{2(out)}/CH_{4(in)}$; H_{wgs} – the enthalpy of the water-gas shift reaction.

According to the results of thermodynamic analysis in Fig.5, the contribution of the WGS reaction to the heat and material balance is insignificant because the change in the Gibbs free energy is minimal compared to the change in the Gibbs free energy for reactions SMR and DMR. Therefore, the equation (10) can be simplified as:

$$H_{overall} \approx X_{CH_4} \cdot H_{mr}. \quad (11)$$

The value of the reaction enthalpy H_{mr} due to the close values of the change in the Gibbs free energy for reactions SMR and DMR can be estimate as follows:

$$H_{mr} = \nu_{H_2O} \cdot H_{smr} + \nu_{CO_2} \cdot H_{DMR}. \quad (12)$$

where ν_{H_2O} and ν_{CO_2} – the mole fraction of H_2O and CO_2 , respectively, in the oxidizer.

$$\nu_{H_2O} = \frac{H_{2O(in)}}{H_{2O(in)} + CO_{2(in)}}; \nu_{CO_2} = \frac{CO_{2(in)}}{H_{2O(in)} + CO_{2(in)}} \quad (13)$$

Therefore, based on the data from Table 2, it is possible to determine the amount of exhaust gas heat, which is transformed into the chemical energy of a new synthetic fuel for various length of the packed beds [13, 17].

Moreover, once the correlation coefficients in Table 2 have been established, the rates of the methane conversion at various operational parameters (such as temperature and inlet reaction mixture composition) can be determined by differentiating function (eq.8) in respect to the residence time W_{cat}/F_{CH_4} . Therefore, the methane conversion rate can be calculated for each residence time as follows:

$$r_{CH_4} = \frac{dX_{CH_4}}{d\left(\frac{W_{cat}}{F_{CH_4}}\right)} = a_1 + 2a_2 \left(\frac{W_{cat}}{F_{CH_4}}\right) + 3a_3 \left(\frac{W_{cat}}{F_{CH_4}}\right)^2 \quad (14)$$

The determination of reaction rate for the methane disappearance in combined methane reforming process with combustion products and steam in this study is determined experimentally. On the other hand, the reaction rate can be determined using a kinetic mechanism as follows:

$$r_{CH_4} = R_{SMR} + R_{DMR} \quad (15)$$

But to develop a generalised kinetic model that can be used for various catalysts shapes and compositions with a change in parameters to suit each catalyst is very difficult issue, if indeed possible. The reasons for this are twofold. First, the change of catalyst composition, size and shape changes not only the values of the parameters of the kinetic model, but it also changes the structure of the kinetic model through changes in the mechanism [47]. Second, a lack of appreciation of the diffusion limitation often results in misunderstandings of the kinetic mechanism [48]. However, it can be argued that the characteristic change in the rate of the reforming reaction is non-linear. The maximum rate of the reforming reaction takes place in the initial section of the reformer, and the minimum rate of the reforming in the section near the outlet. This fact is in good agreement with the work of researchers who have studied the reforming process on commercial nickel catalysts: methane reforming over 3/16 inch pellets of a Ni/alumina catalyst [49], recirculation reactor with a Ni-commercial catalyst of size 0.25-0.63mm [50], commercial catalysts with nickel supported by different materials (in powder form) [51], commercial Ni/MgAl₂O₄ catalyst [24]. Therefore, in order to obtain the best understanding of the methane reforming process on a specified commercial catalyst (Ni- α -Al₂O₃), the experimental study in the present work has been carried out to determine the effect of temperature, pressure and steam addition on the reforming characteristics, in particular, the methane conversion.

Usually, the rate of methane reforming reactions is determined on 1 kg of

catalyst ($\text{mol}/(\text{kg}_{\text{cat}} \cdot \text{s})$). However, as shown by the results of an experimental study, the catalyst shape has a noticeable effect on the methane conversion and as a consequence on the reaction rate (Fig.11). To determine the effect of the catalyst shape on the CH_4 reforming, two catalyst shapes were tested (cylinder 25x25mm with 7 holes with a diameter 5mm; cylinder 15x15mm with 7 holes with a diameter 3mm).

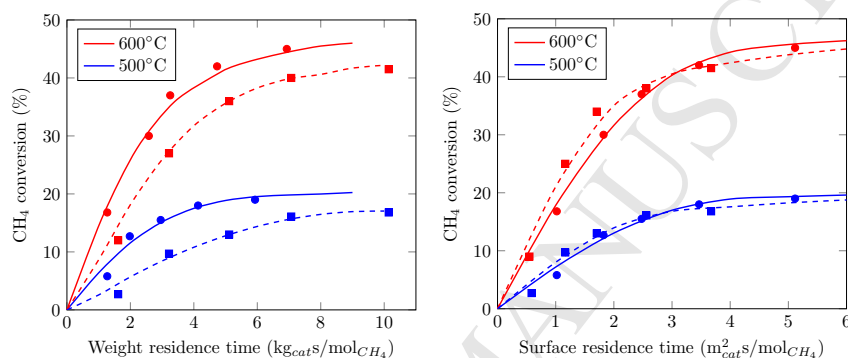


Figure 11: CH_4 conversion vs. residence time: solid line – 15x15mm cylinder; dashed line – 25x25mm cylinder

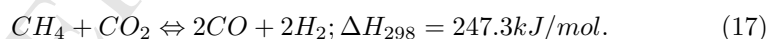
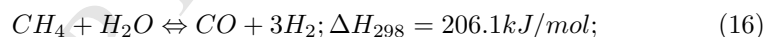
As shown in Fig.11, the experimental methane conversion in a packed bed filled with 7-holes cylinder (15x15mm) is increasing faster than CH_4 conversion in a packed bed filled with 7-holes cylinder (25x25mm) for weight residence time. For example, for the weight residence time of $2 \text{ kg}_{\text{cat}}/\text{s}/\text{mol}$, CH_4 conversion obtained for the 7-holes cylinder (15x15mm) is 1.5 times large than CH_4 conversion obtained for the 7-holes cylinder (25x25mm). In this case, with a further increase in weight residence time, the discrepancy between CH_4 conversions decreases. At the weight residence time above $8 \text{ kg}_{\text{cat}}/\text{s}/\text{mol}$, the methane conversion approaches each other. This can be explained by the fact that with a high weight residence time, the methane reforming processes reaches a near-equilibrium condition. It is known that the catalyst shape does not affect the equilibrium composition of the steam methane reforming reaction.

The methane conversion versus surface residence time is shown in the right

side of Fig.11. The characteristic curve of a dependence of the CH_4 conversion on the surface residence time for the 7-holes cylinder (15x15mm) and the 7-holes cylinders (25x25mm) almost coincide in the range from 0 to $6 \text{ m}_{cat}^2 \text{ s/mol}$. Comparing the left and right parts of Fig.11, it is can be concluded that the catalyst size and shape are the limiting factor for achieving a given CH_4 conversion. This is due to the fact that the reaction majority of the reaction occurs in a region 0.2-0.5mm of the radius from the particle surface (for the industrial Haldor Topsoe R-67-7H catalyst) [52]. Therefore, a direct comparison of the results of this experimental study with the results of other authors is a difficult task, since different shapes, sizes and structures of of the catalysts were applied.

4.3. Effect of pressure

According to the Le Chatelier-Brown principle, the methane reforming process with steam and carbon dioxide is affected not only by the feed gas composition and the temperature but also by the total pressure. This is due to the fact that both steam and carbon dioxide reforming of methane proceeds with an increase in the number of moles of the reaction products. It is can be seen from chemical equations of the steam methane reforming and dry methane reforming. For 2 mole of the inlet reaction mixture, the products of reactions are containing 4 moles.



Using a leakproof burner with a tangential supply of methane allows obtaining the methane combustion at various pressures. Therefore, the effect of pressure on the methane reforming process has been studied.

To study the effect of pressure, the methane reforming process has been investigated for the high residence time about $4.0 \text{ kg}_{cat} \cdot \text{s/mol}$ at the pressure of 1, 3 and 5 bar for the stoichiometric methane-to-oxidant ratio $\text{CH}_4:\text{CO}_2:\text{H}_2\text{O}:\text{N}_2=1:0.33:0.67:2.51$. Fig.12 shows the effect of pressure in the reformer inlet on the methane conversion as a function of temperature when a fixed $\text{CH}_4:\text{CO}_2:\text{H}_2\text{O}:\text{N}_2=1:0.33:0.67:2.51$

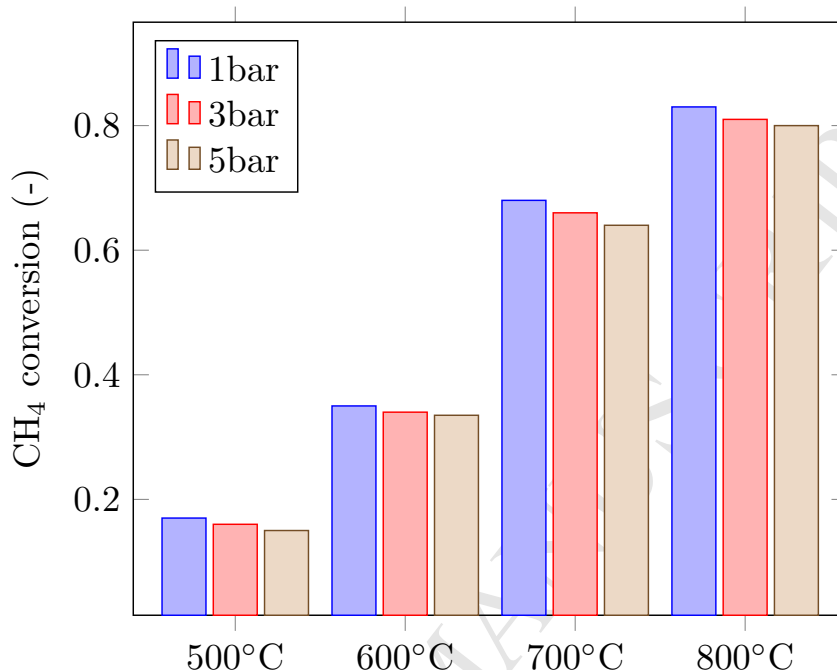


Figure 12: Effect of pressure on the methane conversion at the stoichiometric methane-to-oxidant ratio for the various temperatures

ratio is used. Fig.12 presents that the methane conversion is suppressed by increasing pressure within the temperature range under study. With increasing pressure, the methane conversion decreases because the total product mole number is larger than that of the inlet reactants. In addition, as shown by Xu and Froment, the rate of the conversion reaction is inversely proportional to the pressure [24]. However, the effect of pressure is not more than 1-1.8% (for the investigated pressure range from 1 to 5 bar) and in engineering calculations, the effect of pressure at a moderately low pressure (below 5 bar) can be considered insignificant. The substitution effect of pressure on CH₄ conversion takes place at a higher pressure [53].

4.4. Catalyst activity

However, in practice, the use of methane-to-oxidant ratio equal to 1.0 is difficult, because even with an experiment duration of about 15 minutes, the traces of soot carbon are observed on the surface of the catalyst after experiments. However, the gas analyzer Gasboard 3100 is allowing to analyze the composition of only the gaseous components of the reforming products. In addition, it is known that the formation of soot depends on the activity of the catalyst [26]. To evaluate the activity of the catalyst, a series of experiments was carried out, when the experiment was performed for about 10 hours. These results are shown below in Fig.14, while the composition of the reforming products is shown in Fig.13. If the catalyst activity is considered to be the criterion of optimality, then the optimal amount of steam is $H_2O=1.0$.

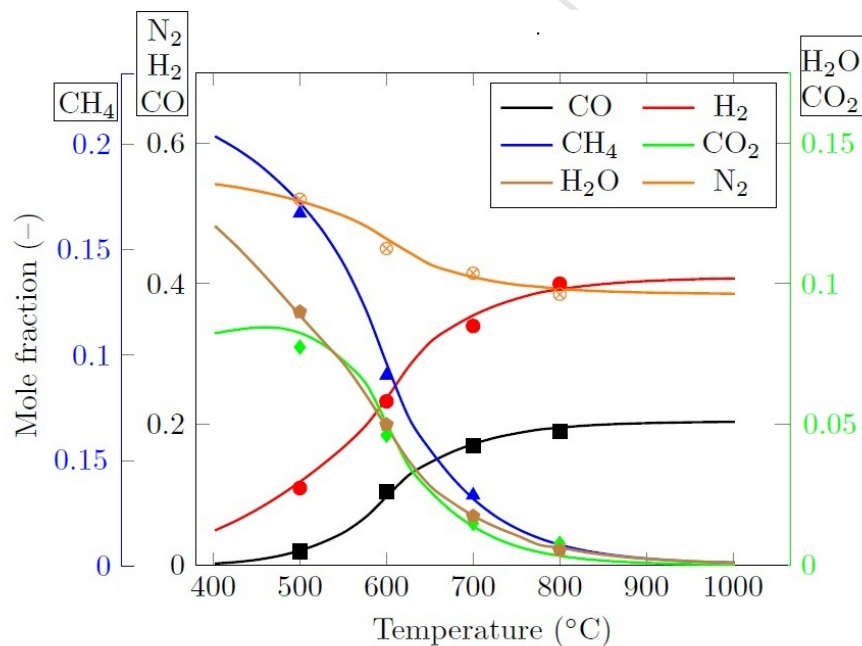


Figure 13: Composition of reforming products as a function of temperature for a stoichiometric methane-to-oxidant ratio at $p=1\text{bar}$: solid line – equilibrium composition; points – experimental results

Fig.13 depicts the variation of the product gas concentrations as a func-

tion of reforming temperature for the stoichiometric methane-to-oxidant ratio of $\text{CH}_4:\text{CO}_2:\text{H}_2\text{O}:\text{N}_2=1:0.33:0.67:2.51$. It is clear that at higher reforming temperature the methane residual in the product gas is lower and the hydrogen concentration is higher. Experimental data and results of the thermodynamic analysis have a good correlation.

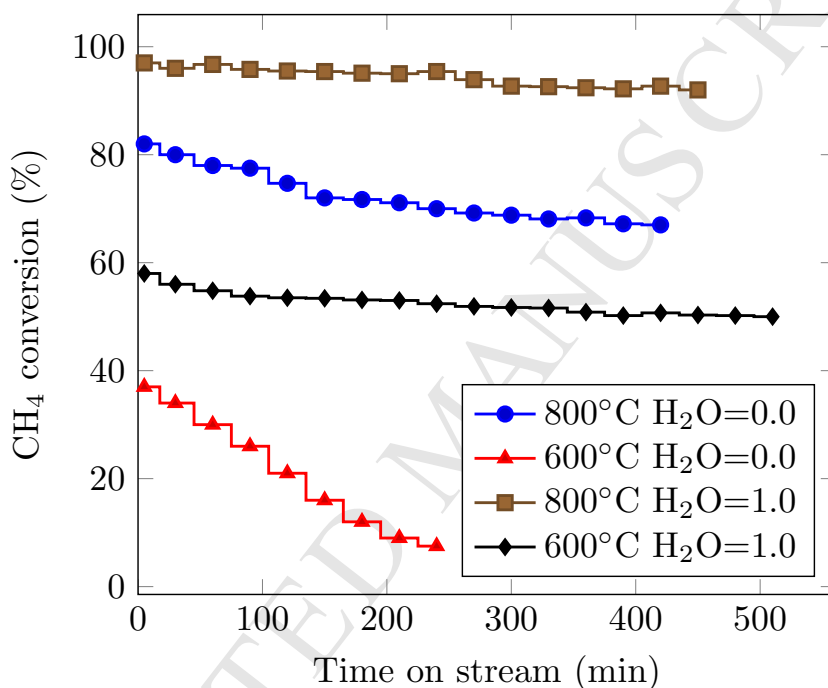


Figure 14: Variation of the methane conversion with time for a fixed stoichiometric methane-to-oxidant ratio at $p=1\text{bar}$ when various mole number of steam is added

Fig.14 shows the variation of the CH_4 conversion at different conditions with time. Four experiments were performed at 200–500 min on stream. For a stoichiometric methane-to-oxidant ratio the methane conversion decreases with decreasing of temperature. A decrease in the methane conversion occurs due to the catalyst deactivation. The catalyst deactivation might be due to carbon formation in the catalyst particle. Based on a thermodynamic analysis of possible carbon formation reactions (Table 1), it is clear that the catalyst deactivation

resulted mainly from the reactions of methane cracking, Boudouard reaction, and carbon gasification. From the analysis of the chemical equations shown in Table 1, it is seen that in order to prevent the carbon formation, the following conditions must be fulfilled:

– for temperature above $\sim 950\text{K}$:

$$\frac{P_{H_2}^2}{P_{CH_4}} > K_{eq,4}; \quad \frac{P_{CO_2}}{P_{CO}} > K_{eq,5}; \quad (18)$$

– for temperature below $\sim 950\text{K}$:

$$\frac{P_{CO} \cdot P_{H_2}}{P_{H_2O}} < K_{eq,6}; \quad \frac{P_{CO_2} \cdot P_{H_2}^2}{P_{H_2O}^2} < K_{eq,7}; \quad (19)$$

where i -index for the equilibrium constants $K_{eq,i}$ is number of reaction in Table 1.

According to Fig.14, the methane conversion drops during the first 200 min for all conditions. However, the CH_4 conversion for high temperature and high amount of steam in the reaction mixture decreases more slowly when for low temperature and low amount of extra steam. Fig.14 shows that moderately high temperatures must be used for the industrial application of the thermochemical recuperation systems. This conclusion was made because a decrease in the catalytic activity will lead to a decrease in the efficiency of the TCR systems. The experimental results in this study fulfil the conditions expressed by Eqs. (9) and (10). This is due to fact that at the high residence time the reaction products achieved close to equilibrium composition.

To inhibit the reaction of carbon formation, it is necessary to add steam to the reaction mixture. However, the addition of steam leads to an increase in the heat spent on its production. Moreover, extra steam in the reformat products reduces the specific lower heat value (LHV) of synthesis gas. These facts are important for thermochemical recovery systems from the point of view of energy efficiency, and the problem of determining the optimal methane-to-oxidant ratio may be the subject of a separate study. However, it is known that the hydrogen addition to the initial reaction mixture can significantly reduce the deposition of carbon on the catalyst surface [26, 54, 55]. Therefore, in order

to increase the activity of the catalyst, hydrogen might be added to the initial reaction mixture. The set of experiments to investigate the effect of carbon deposition on the reforming characteristics where hydrogen is added to inlet reaction mixture will be conducted in future author's papers.

5. Conclusion

An experimental study and thermodynamic analysis of the methane reforming process with products of methane combustion and steam over a commercial Ni-based catalyst has been carried out on a commercial Ni-based catalyst in the wide ranges of operational parameters such as temperature, residence time, inlet gas composition and pressure. The thermodynamic analysis was carried out at thermodynamic equilibrium by total Gibbs free energy minimization of the reforming system to determine the maximum achievable methane conversion in the reformer. An addition of steam to flue gas increases of the methane conversion. The methane conversion reaches the value that is close to equilibrium at the residence time about $4.29 \text{ kg}_{cat}\cdot\text{s}/\text{mol}$ for all investigated inlet gas compositions. The methane conversion is nearly directly proportional to residence time at the temperature range lower than 700°C . For temperatures above 800°C , the methane conversion at a low residence time increases faster than for a high contact time. With increasing pressure, the methane conversion decreases. However, the effect of pressure is not more than 1-1.8% (for the investigated pressure range from 1 to 5 bar) and in engineering calculations, the effect of pressure at a moderately low pressure (below 5 bar) can be considered insignificant. For a stoichiometric methane-to-oxidant ratio the methane conversion decreases with decreasing of temperature with increasing time on stream. The CH_4 conversion for temperature above 700°C with extra steam in the reaction mixture decreases more slowly when for low temperature and low amount of extra steam for time on stream above 200 min.

Acknowledgment

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Methane reforming with a mixture of methane combustion products and steam

The methane conversion close to equilibrium above $4\text{kg}_{\text{cat}}\cdot\text{s}/\text{mol}$

Addition of steam to flue gas increase the methane conversion

The effect of pressure (in the range from 1 to 5 bar) on methane conversion is insignificant

The catalyst activity decreases rapidly for the temperature range below 700°C and the stoichiometric methane-to-flue gas ratio

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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